Shifted-Chart Composition for Horizon-Optimal Time-Varying LQR

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***Abstract*— Horizon optimization in finite-horizon LQR treats the horizon T as a decision variable and adds a penalty wT to the cost. Recent composition-based methods evaluate all candidate horizons in $O(Nn^3)$, instead of the $O(N^2n^3)$ of one Riccati recursion per horizon, but they track the inverse cost-to-go P^{-1} and therefore require every stage and terminal cost to be positive definite, whereas LQR requires only positive semidefiniteness. This excludes costs that leave some states unpenalized. In this work, we track instead the shifted inverse $\widehat{P}_k = (P_k + cI)^{-1}$, the inverse coordinate after a symplectic shear of the state–costate variables. Its domain contains every positive semidefinite Riccati matrix, and $P_k = \widehat{P}_k^{-1} - cI$ recovers the Riccati solution. We show that the shift can be absorbed into the stage coefficients of the inverse-coordinate composition, so the $O(Nn^3)$ search carries over, exact in exact arithmetic, subject to invertibility conditions. These hold for all small c when the dynamics contract the unpenalized directions, fail for at most finitely many c outside a degenerate case, and are certified per horizon at run time. In our experiments, diagonal loading $Q_k + \varepsilon I$ either left cost errors several orders of magnitude above those of the shifted chart or, for small fixed ε, selected the wrong horizon. The shifted chart with a fixed default c selected the correct horizon on all tested instances, reporting long horizons as uncertified in one stress case.**

Nomenclature

Notation	
$(X) \succeq 0, (X) \succ 0$	$(\bullet)$ being symmetric positive semidefinite, positive definite matrix
Horizon-Optimal LQR Parameters	
$x_k,\ u_k$	state and input
$A_k,\ B_k$	time-varying dynamics
$Q_k \succeq 0,\ Q_T \succeq 0$	stage and terminal state costs
$R_k \succ 0$	input cost
D_k	$B_k R_k^{-1} B_k^\top$
$T,\ N,\ T^\star$	horizon, maximum horizon, optimal horizon
$w,\ J_T$	horizon penalty, optimal cost for horizon T
$\mathcal{N}(Q_k),\ \Pi_k$	unpenalized subspace, its orthogonal projector
Coordinates	
$P_k,\ \lambda_k$	Riccati matrix, costate ($\lambda_k = P_k x_k$)
$\mathcal{L}_k$	state–costate subspace $\mathrm{col}[I,\ P_k]^\top$
$\widetilde{P}_k$	inverse coordinate P_k^{-1} [1]
$\widehat{P}_k$	shifted inverse coordinate $(P_k + cI)^{-1}$
$c,\ c_0$	shift, default shift
ε	diagonal loading, $Q_k \mapsto Q_k + \varepsilon I$
Composition	
E_k, F_k, G_k	inverse-coordinate stage coefficients
$\widehat{E}_k, \widehat{F}_k, \widehat{G}_k$	shifted stage coefficients
E'_k, F'_k, G'_k	absorbed coefficients (Lemma 1)
$\bar{E}_k, \bar{F}_k, \bar{G}_k$	composed coefficients of stages $0, \ldots, k$
M_k	$I - c\widehat{G}_k$, pivot (a)
$\phi,\ h_k,\ \psi_k$	unshift map, shifted stage map, $\psi_k = h_k \circ \phi$
Certification	
$\tau,\ \mathcal{C}$	rcond threshold, certified horizons

I. INTRODUCTION

Optimal control problems (OCPs) seek a trajectory of states and controls that minimizes a cost functional subject to the dynamics. Common approaches such as LQR [2], MPC [3], iLQR [4] and DDP [5] use a fixed planning horizon. In many applications, however, a shorter horizon reduces computation while a longer horizon lowers the task cost, which has motivated treating the horizon length as a decision variable of the OCP [6]–[8], including recent schemes that adapt the horizon online from learned horizon maps [9].

For linear quadratic problems, finite-horizon LQR solves a fixed horizon by a backward Riccati recursion [10]. A brute-force search over $T = 1, \ldots, N$ repeats overlapping portions of this recursion, giving an N^2 dependence on the maximum horizon N. For stationary (A, B, Q, R), [11] exploits the fact that the per-step map does not depend on the horizon: a single backward pass computes all Riccati matrices, and the cost for any T is read off by re-indexing. This property fails for time-varying data, which is the generic case when nonlinear dynamics are linearized along a trajectory.

A geometric view treats the Riccati matrix as a coordinate on a space of subspaces. For a fixed horizon, the costate λ_k is the gradient of the cost-to-go $V_k(x) = \frac{1}{2} x^\top P_k x$, so along optimal trajectories $\lambda_k = P_k x_k$. The state–costate pairs satisfying this relation form the n-dimensional subspace $\mathcal{L}_k = \mathrm{col}[I,\ P_k]^\top \subset \mathbb{R}^{2n}$, the column space of the $2n \times n$ matrix $[I,\ \ P_k]^\top$, also called the graph of P_k (for the algebraic equation, the graph of an invariant subspace of a symplectic pencil [12]). It is a point of the Grassmannian $\mathrm{Gr}(n, 2n)$, the set of all n-dimensional subspaces of $\mathbb{R}^{2n}$. The Riccati recursion propagates $\mathcal{L}_k$ backward in time through $\mathrm{Gr}(n, 2n)$, and P_k is a local coordinate for it, a *graph coordinate*; see [13] for the continuous-time analogue. The same subspace admits other graph coordinates: when P_k is invertible, $\mathcal{L}_k = \mathrm{col}[P_k^{-1},\ I]^\top$, the graph of $\widetilde{P}_k = P_k^{-1}$ over the costate. Under a linear change of the state–costate variables, a graph coordinate undergoes a matrix linear fractional transformation (LFT). The composition of [1] uses the inverse coordinate $\widetilde{P}_k$: in it, consecutive time-varying Riccati maps compose into a single LFT, so all candidate horizons can be queried after one forward composition pass, at $O(Nn^3)$ cost.

The inverse coordinate requires Q_k^{-1} and Q_T^{-1}, whereas

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Submitted to the 2027 American Control Conference (ACC); under review.
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LQR requires only $Q_k, Q_T \succeq 0$. Singular costs arise whenever some states are not penalized. Diagonal loading, $Q_k \mapsto Q_k + \varepsilon I$, restores invertibility but changes the problem being solved, and with it the optimal cost, the control law and possibly the selected horizon. Alternatively, lifting the problem over d steps yields a positive definite cost under uniform d-step observability and nonsingular A_k [14]; the shifted coordinate requires neither assumption and leaves the stage structure unchanged.

The contributions of this work are:

1) *Shifted inverse coordinate.* We represent the Riccati matrix by $\widehat{P}_k = (P_k + cI)^{-1}$, the inverse coordinate after a fixed symplectic shear, whose domain contains every positive semidefinite Riccati matrix (Proposition 1).
2) *Absorption into the composition.* We show that the coordinate change can be absorbed into the stage coefficients of the inverse-coordinate LFT, so that horizon search keeps its $O(Nn^3)$ cost and, in exact arithmetic, recovers the Riccati solution exactly, provided the invertibility conditions of Assumption 1 hold. These conditions hold for all small c when the dynamics contract the unpenalized directions, fail for at most finitely many c outside a degenerate case, and are certified per horizon at run time.
3) *Numerical comparison.* On random time-varying instances and three physical systems with singular Q_k, we compare against diagonal loading, an adaptive-jitter inverse coordinate and the released HOP implementation, and report where each fails, including a stress case in which the shifted chart reports some horizons as uncertified.

Section II states the problem and reviews the inverse-coordinate composition, Section III develops the shifted chart and its admissibility, Section IV reports the numerical study, and Section V provides concluding remarks.

II. Preliminaries and Problem Formulation

A. Horizon-Optimal Time-Varying LQR

Consider the discrete-time linear time-varying system

$$x_{k+1} = A_k x_k + B_k u_k, \qquad k = 0, \dots, T-1, \tag{1}$$

where $x_k \in \mathbb{R}^n$ and $u_k \in \mathbb{R}^m$, and A_k, B_k are of compatible dimensions. The horizon T is selected from

$$T \in \{1, \dots, N\}, \tag{2}$$

where N denotes the maximum allowable horizon.

For a candidate horizon T, consider the quadratic objective

$$J_T = \frac{1}{2} x_T^\top Q_T x_T + \frac{1}{2} \sum_{k=0}^{T-1} \left(x_k^\top Q_k x_k + u_k^\top R_k u_k \right) + wT, \tag{3}$$

where $w \geq 0$ penalizes the horizon length. We assume

$$R_k \succ 0, \tag{4}$$

while $Q_k, Q_T \succeq 0$ need not be invertible; Q_k, Q_T are $n \times n$ and R_k is $m \times m$.

The horizon-optimal LQR problem is then

$$T^\star = \operatorname*{arg\,min}_{1 \leq T \leq N} J_T, \tag{5}$$

where J_T denotes the optimal value of (3) for a given candidate horizon T and fixed initial state x_0.

B. Finite-Horizon Riccati Recursion

For a fixed candidate horizon T, the finite-horizon LQR problem can be solved by dynamic programming. The optimal value function is quadratic,

$$V_k(x_k) = \frac{1}{2} x_k^\top P_k x_k, \tag{6}$$

with terminal condition

$$P_T = Q_T. \tag{7}$$

The backward Riccati recursion is given by

$$S_k = R_k + B_k^\top P_{k+1} B_k, \tag{8}$$

$$K_k = S_k^{-1} B_k^\top P_{k+1} A_k, \tag{9}$$

$$P_k = Q_k + A_k^\top P_{k+1} A_k - A_k^\top P_{k+1} B_k K_k. \tag{10}$$

The corresponding optimal control law is

$$u_k^\star = -K_k x_k. \tag{11}$$

The optimized cost for horizon T is consequently

$$J_T = \frac{1}{2} x_0^\top P_0 x_0 + wT. \tag{12}$$

The Riccati recursion (10) does not require Q_k^{-1}: since $R_k \succ 0$, the matrix S_k in (8) is positive definite, and if $Q_k, Q_T \succeq 0$, then $P_k \succeq 0$ for all k. The fixed-horizon recursion is therefore well defined even when Q_k is singular.

C. Horizon Search and Computational Complexity

A direct approach to (5) solves the finite-horizon LQR problem independently for every candidate horizon $T = 1, \dots, N$. The candidate horizon T requires T backward Riccati steps, resulting in

$$\sum_{T=1}^{N} T = \mathcal{O}(N^2) \tag{13}$$

Riccati updates. Since each update requires $\mathcal{O}(n^3)$ matrix operations, the overall complexity of the brute-force horizon search is

$$\mathcal{O}(N^2 n^3). \tag{14}$$

D. LFT Composition in the Inverse Coordinate

Following HOP [1], define the inverse cost-to-go

$$\widetilde{P}_k = P_k^{-1}. \tag{15}$$

Applying the Woodbury Matrix Identity[15] to the Riccati recursion10 gives

$$\widetilde{P}_k = E_k - F_k \left(\widetilde{P}_{k+1} + G_k \right)^{-1} F_k^\top, \tag{16}$$

where

$$E_k = Q_k^{-1}, \tag{17}$$
$$F_k = Q_k^{-1} A_k^\top, \tag{18}$$
$$G_k = A_k Q_k^{-1} A_k^\top + B_k R_k^{-1} B_k^\top. \tag{19}$$

Thus, each Riccati step can be represented by the map

$$\widetilde{g}_k(X) = E_k - F_k (X + G_k)^{-1} F_k^\top. \tag{20}$$

The maps in (20) are closed under composition. In particular, the composition of successive Riccati maps can be represented using another triple of matrices,

$$\widetilde{g}_{0:k}(X) = \bar{E}_k - \bar{F}_k (X + \bar{G}_k)^{-1} \bar{F}_k^\top. \tag{21}$$

Starting from

$$(\bar{E}_0, \bar{F}_0, \bar{G}_0) = (E_0, F_0, G_0), \tag{22}$$

the composed parameters can be updated recursively as

$$W_k = (E_k + \bar{G}_{k-1})^{-1}, \tag{23}$$
$$\bar{E}_k = \bar{E}_{k-1} - \bar{F}_{k-1} W_k \bar{F}_{k-1}^\top, \tag{24}$$
$$\bar{F}_k = \bar{F}_{k-1} W_k F_k, \tag{25}$$
$$\bar{G}_k = G_k - F_k^\top W_k F_k. \tag{26}$$

For a candidate horizon T, the terminal inverse cost is

$$\widetilde{P}_T = Q_T^{-1}, \tag{27}$$

and the initial inverse cost-to-go can be obtained from the composed map as

$$\widetilde{P}_0^{(T)} = \widetilde{g}_{0:T-1} \left(Q_T^{-1} \right). \tag{28}$$

The corresponding horizon cost is recovered as

$$J_T = \frac{1}{2} x_0^\top \left(\widetilde{P}_0^{(T)} \right)^{-1} x_0 + wT. \tag{29}$$

The composed maps can be constructed in a single forward pass over the horizon. Consequently, the horizon-dependent Riccati information can be reused rather than recomputed independently for every candidate horizon, giving an overall complexity of

$$\mathcal{O}(Nn^3), \tag{30}$$

compared with the $\mathcal{O}(N^2 n^3)$ complexity of the brute-force approach.

The coefficients (E_k, F_k, G_k) and the terminal query require Q_k^{-1} and Q_T^{-1}, which excludes singular state costs. The next section introduces a shifted inverse coordinate that removes this requirement while recovering the Riccati matrix exactly.

III. Shifted-Chart Composition

The inverse coordinate of Section II-D requires Q_k^{-1} and Q_T^{-1}, although LQR permits $Q_k \succeq 0$. Rather than altering the cost by diagonal loading, we change the coordinate in which the state-costate subspace is represented, keeping the three-parameter LFT structure that composition requires.

A. Shifted Inverse Coordinate

The state-costate subspace associated with a Riccati matrix can be written as the graph

$$\mathcal{L}_k = \operatorname{col} \begin{bmatrix} I \\ P_k \end{bmatrix}. \tag{31}$$

When P_k is nonsingular, the same subspace can equivalently be represented as

$$\mathcal{L}_k = \operatorname{col} \begin{bmatrix} P_k^{-1} \\ I \end{bmatrix}, \tag{32}$$

which corresponds to describing the subspace as a graph over the costate, $x_k = P_k^{-1} \lambda_k$. Thus, the inverse Riccati coordinate used in Section II-D is a coordinate representation of the same state-costate subspace, but it is defined only when P_k is invertible.

We instead apply a fixed shear to the state-costate variables,

$$S_c : (x, \lambda) \mapsto (x, \lambda + cx), \qquad c > 0, \tag{33}$$

with matrix representation

$$S_c = \begin{bmatrix} I & 0 \\ cI & I \end{bmatrix}. \tag{34}$$

Since cI is symmetric, S_c is symplectic. Its action on the Riccati graph is

$$S_c \begin{bmatrix} I \\ P_k \end{bmatrix} = \begin{bmatrix} I \\ P_k + cI \end{bmatrix}. \tag{35}$$

Taking the inverse graph coordinate in these sheared coordinates gives the shifted inverse coordinate

$$\hat{P}_k := (P_k + cI)^{-1}, \qquad P_k = \hat{P}_k^{-1} - cI. \tag{36}$$

Thus, the proposed coordinate is the inverse coordinate applied after a fixed symplectic shear of the state-costate variables. The underlying Riccati matrix, and hence the original LQR cost-to-go is recovered exactly.

Proposition 1 (Chart admissibility)**.** *Every $P \succeq 0$ lies in the domain of $\widehat{P} = (P + C)^{-1}$ if and only if $C \succ 0$. In particular, for $C = cI$ with $c > 0$, all Riccati iterates P_k and all $Q_k + cI$, $Q_T + cI$ are invertible, with $\lambda_{\min}(P_k + cI) \geq c$.*

Proof. If $C \succ 0$, then $v^\top (P + C) v \geq v^\top C v > 0$ for $v \neq 0$. Conversely, if $Cv = \lambda v$ with $v \neq 0$ and $\lambda \leq 0$, then $P = -\lambda v v^\top \succeq 0$ and $(P + C)v = 0$. □

B. Shifted Stage Map and LFT Absorption

For invertible P_{k+1}, write $\widetilde{P}_{k+1} = P_{k+1}^{-1}$. The Woodbury identity rewrites the Riccati recursion (10) as $P_k = Q_k + A_k^\top \big(\widetilde{P}_{k+1} + B_k R_k^{-1} B_k^\top \big)^{-1} A_k$. Adding cI to both sides and inverting once more with Woodbury gives

$$\widehat{P}_k = h_k(\widetilde{P}_{k+1}) := \widehat{E}_k - \widehat{F}_k \big(\widetilde{P}_{k+1} + \widehat{G}_k \big)^{-1} \widehat{F}_k^\top, \tag{37}$$

with stage coefficients that require only $Q_k + cI \succ 0$:

$$\widehat{E}_k = (Q_k + cI)^{-1}, \tag{38}$$
$$\widehat{F}_k = \widehat{E}_k A_k^\top, \tag{39}$$
$$\widehat{G}_k = A_k \widehat{E}_k A_k^\top + B_k R_k^{-1} B_k^\top. \tag{40}$$

The map h_k consumes the unshifted inverse $\widetilde{P}_{k+1}$, whereas the recursion carries $\widehat{P}_{k+1}$. By (36) the two are related by

$$\widetilde{P} = \big(\widehat{P}^{-1} - cI\big)^{-1} = \widehat{P}\big(I - c\widehat{P}\big)^{-1} =: \phi(\widehat{P}), \qquad (41)$$

so the one-step update in shifted coordinates is $\psi_k := h_k \circ \phi$. The following lemma absorbs ϕ into the stage coefficients.

Lemma 1 (Absorption)**.** *Let $\mathcal{R}_k$ denote the Riccati map* (10)*, $P_k = \mathcal{R}_k(P_{k+1})$, and define the exact shifted step*

$$\psi_k(X) := \big(\mathcal{R}_k(X^{-1} - cI) + cI\big)^{-1},\ X = (P + cI)^{-1},\ P \succeq 0. \qquad (42)$$

Assume $M_k := I - c\widehat{G}_k$ is invertible and let

$$E'_k = \widehat{E}_k + c\,\widehat{F}_k M_k^{-1} \widehat{F}_k^{\top}, \qquad (43)$$

$$F'_k = \widehat{F}_k M_k^{-1}, \qquad (44)$$

$$G'_k = \widehat{G}_k M_k^{-1}. \qquad (45)$$

Then E'_k and G'_k are symmetric, and

$$\psi_k(X) = E'_k - F'_k\,(X + G'_k)^{-1} {F'_k}^{\top} \qquad (46)$$

for every X in (42) *with $X + G'_k$ invertible.*

Proof. Drop the index k and let $M = I - c\widehat{G}$. As M is a polynomial in the symmetric $\widehat{G}$, M^{-1} is symmetric and commutes with $\widehat{G}$, so E' and G' are symmetric. First let $P \succ 0$; by the derivation above, $\psi = h \circ \phi$. Since X commutes with $(I - cX)^{-1}$ and $\widehat{G} = \widehat{G}M^{-1}M$,

$$\begin{aligned}\phi(X) + \widehat{G} &= (I - cX)^{-1}\big[X + (I - cX)\widehat{G}\big] \\ &= (I - cX)^{-1}(X + G')M.\end{aligned}$$

Since $M^{-1} - cG' = (I - c\widehat{G})M^{-1} = I$, we have $I - cX = M^{-1} - c(X + G')$, so

$$\big(\phi(X) + \widehat{G}\big)^{-1} = M^{-1}(X + G')^{-1}M^{-1} - cM^{-1}.$$

Substituting into h and using $M^{-1}\widehat{F}^{\top} = {F'}^{\top}$,

$$\begin{aligned}\psi(X) &= \widehat{E} + c\widehat{F}M^{-1}\widehat{F}^{\top} - F'(X + G')^{-1}{F'}^{\top} \\ &= E' - F'(X + G')^{-1}{F'}^{\top}.\end{aligned}$$

For singular $P \succeq 0$, let $P_j = P + I/j \succ 0$ and $X_j = (P_j + cI)^{-1} \to X$. The left side $\psi(X_j) = (\mathcal{R}(P_j) + cI)^{-1}$ converges to $\psi(X)$, since $\mathcal{R}$ is continuous on $P \succeq 0$ (as $R \succ 0$) and $\mathcal{R}(P) + cI \succeq cI$. The right side converges because $X + G'$ is invertible. Letting $j \to \infty$ gives the identity at P. □

Remark 1 (Matrix Shifts)**.** *Lemma 1 extends to symmetric shears $S_C = \begin{bmatrix} I & 0 \\ C & I \end{bmatrix}$ with symmetric C, giving $E' = \widehat{E} + \widehat{F}C(I - \widehat{G}C)^{-1}\widehat{F}^{\top}$, $F' = \widehat{F}(I - C\widehat{G})^{-1}$, and $G' = (I - \widehat{G}C)^{-1}\widehat{G}$. We chose $C = cI$ to avoid non-commutative factor ordering and reduce parameter tuning to scalar c.*

C. Admissibility and Choice of the Shift

Proposition 1 guarantees that the shifted *chart* contains every Riccati matrix for any $c > 0$. It does not guarantee that the stage *maps* keep the three-parameter form with finite coefficients, nor that their composition and the horizon queries do. Each condition below is the pivot of a composition in family (20): of h_k with ϕ for (a), of the prefix with stage k for (b), and of the prefix with the terminal value for (c).

Assumption 1 (Admissibility)**.** *For a given c: (a) $I - c\widehat{G}_k$ is invertible for all k; (b) $E'_k + \bar{G}_{k-1}$ is invertible for all k; (c) $(Q_T^{(t)} + cI)^{-1} + \bar{G}_{t-1}$ is invertible for all t.*

For $Q_k, Q_T \succeq 0$, the shifted coordinate and $\widehat{E}_k$ exist for every $c > 0$ (Proposition 1); only the stage maps and their composition can fail, and they can fail for every c (Remark 2).

Theorem 2 (Shifted-chart horizon search)**.** *Let $Q_k, Q_T \succeq 0$, $R_k \succ 0$ and $c > 0$, and let Assumption 1 hold. Then the triples (E'_k, F'_k, G'_k) compose by* (23)–(26) *with $(E_k, F_k, G_k) \to (E'_k, F'_k, G'_k)$, and Algorithm 1 returns, in exact arithmetic, the optimal cost J_T for every $T \in \{1, \dots, N\}$ in $O(Nn^3)$ operations, without inverting Q_k or Q_T and without modifying the cost.*

Proof. By Proposition 1 the coefficients $\widehat{E}_k, \widehat{F}_k, \widehat{G}_k$ exist, and by Lemma 1 each exact shifted step ψ_k, including at singular P_{k+1}, lies in the family (20). The composition rule (23)–(26) uses only this form, so it applies unchanged and gives $\widehat{P}_0^{(T)} = \psi_{0:T-1}\big((Q_T + cI)^{-1}\big)$; then $P_0 = (\widehat{P}_0^{(T)})^{-1} - cI$ by (36). The N stage constructions, $N - 1$ composition steps and N queries each cost $O(n^3)$. □

Proposition 2 (Generic admissibility)**.** *For fixed data, the determinants of the matrices in Assumption 1 are rational functions of c on $c > 0$. Unless one vanishes identically, Assumption 1 fails for at most finitely many $c > 0$.*

Proof. $\widehat{E}_k = (Q_k + cI)^{-1}$ is rational in c with poles only at $-\lambda_i(Q_k) \le 0$, so $\widehat{F}_k, \widehat{G}_k$, and wherever (a) holds E'_k, F'_k, G'_k, are rational on $c > 0$. By induction through (23)–(26), so are $\bar{E}_k, \bar{F}_k, \bar{G}_k$ and the matrix in (c). A nonzero rational function has finitely many zeros. □

Remark 2 (Persistent failure)**.** *If $w \neq 0$ satisfies $B_k^{\top} w = 0$, $Q_k A_k^{\top} w = 0$ and $A_k A_k^{\top} w = w$, then $\widehat{E}_k A_k^{\top} w = A_k^{\top} w/c$ and $(I - c\widehat{G}_k)w = 0$ for every $c > 0$. This is the degenerate case of Proposition 2; it has $\|A_k \Pi_k\|_2 \ge 1$.*

Remark 3 (Factorization)**.** *The pivots are symmetric but, outside the regime of Proposition 3, indefinite. They are factored by LU or $LDL^{\top}$, not Cholesky.*

Proposition 3 (Sufficient condition)**.** *Let $D_k = B_k R_k^{-1} B_k^{\top}$ and let Π_k project onto $\mathcal{N}(Q_k)$. (i) If $c\,\lambda_{\max}(\widehat{G}_k) < 1$ for all k, every pivot is symmetric positive definite. (ii) Such $c > 0$ exists if and only if $\|A_k \Pi_k\|_2 < 1$ for all k; then (i) holds for all $c \in (0, \bar{c})$, with $\bar{c} = \min_k \big(1 - \|A_k \Pi_k\|_2^2\big) / \big(\|A_k(I - \Pi_k)\|_2^2 / \lambda_+(Q_k) + \|D_k\|_2\big)$ and λ_+ the smallest positive eigenvalue.*

Proof. *(i)* Drop the index k and let $M = I - c\widehat{G} \succ 0$ and $S = A\widehat{E}A^\top$, so that $\widehat{G} = S + D$ and $M + cS = I - cD \succ 0$. Then $E' \succ 0$ and $G' = \widehat{G}M^{-1} \succeq 0$. Multiplying E' by $\widehat{F}^\top \widehat{E}^{-1}$ gives $\widehat{F}^\top \widehat{E}^{-1}E' = (I + cSM^{-1})\widehat{F}^\top$, from which

$$F'^\top E'^{-1}F' = (I - cD)^{-1}SM^{-1},$$
$$G' - F'^\top E'^{-1}F' = (I - cD)^{-1}D \succeq 0.$$

If $\bar{G}_{k-1} \succeq 0$, then $E'_k + \bar{G}_{k-1} \succeq E'_k \succ 0$, so $W_k \preceq E'^{-1}_k$ and

$$\bar{G}_k \succeq G'_k - F'^\top_k E'^{-1}_k F'_k \succeq 0.$$

Since $\bar{G}_0 = G'_0 \succeq 0$, this holds for all k, and pivot (c) is positive definite.

(ii) The matrix $c\widehat{E}_k = (I + Q_k/c)^{-1}$ has eigenvalue 1 on $\mathcal{N}(Q_k)$ and $c/(c+\lambda_i) \le c/\lambda_+$ otherwise, so

$$\Pi_k \preceq c\widehat{E}_k \preceq \Pi_k + \tfrac{c}{\lambda_+}(I - \Pi_k).$$

Substituting into $c\widehat{G}_k = A_k(c\widehat{E}_k)A_k^\top + cD_k$ gives

$$c\lambda_{\max}(\widehat{G}_k) \ge \|A_k\Pi_k\|^2,$$
$$c\lambda_{\max}(\widehat{G}_k) \le \|A_k\Pi_k\|^2 + c\big(\|A_k(I-\Pi_k)\|^2/\lambda_+ + \|D_k\|\big),$$

and $c\widehat{G}_k$ is nondecreasing in c. □

Under (i), the shifted maps are inverse-coordinate maps of an auxiliary problem with positive definite cost E'^{-1}_k, i.e., the setting of [1]. The physical systems of Section IV have $\|A_k\Pi_k\|_2 \ge 1$; there the pivots are indefinite (Remark 3), and admissibility rests on Proposition 2 and run-time certification.

Run-time certification. Algorithm 1 estimates the reciprocal condition number of every pivot from its LU factors, at $O(n^2)$ cost each. Horizon t is *certified* if every pivot it uses, namely (a) and (b) at stages $0, \dots, t-1$ and (c) at t, has reciprocal condition number at least τ. Uncertified horizons may be retried with $c \leftarrow \rho c$; a retry can only add certified horizons. The algorithm returns J_t and the minimizer over certified horizons only, with the certified set, so a violation of Assumption 1 is reported rather than returned as a cost, and the $O(Nn^3)$ bound is unchanged. Certification bounds the conditioning of each pivot, not the final error: near the threshold the error can approach u/τ, so τ trades coverage against accuracy.

IV. Numerical Study

Errors are $\max_{1\le t\le N} |J_t - J_t^{\mathrm{ref}}|/\max(|J_t^{\mathrm{ref}}|, 10^{-12})$, the worst disagreement over all candidate horizons, and a horizon is *wrong* when the method's $T^\star$ differs from brute force. Unless stated otherwise, Shifted-HOP runs Algorithm 1 with $c = c_0 = 1$, the checks of Sec. III-C enabled ($\tau = 10^{-10}$), and no tuning. Loading denotes the inverse coordinate with $Q_k \mapsto Q_k + \varepsilon I$; *oracle* ε is the grid value (10^{-20}–10^{2}) minimizing cost error against brute force, which is unavailable in practice. Naive HOP is the inverse coordinate with an escalating Cholesky jitter starting at 10^{-10}.

Algorithm 1 Shifted HOP-LQR with certification

Require: $\{A_k, B_k, Q_k, R_k\}_{k=0}^{N-1}$, $\{Q_T^{(t)}\}_{t=1}^{N}$, x_0, w, c, τ
Ensure: certified set $\mathcal{C}$, $\{J_t\}_{t\in\mathcal{C}}$, $T^\star = \arg\min_{t\in\mathcal{C}} J_t$

Stage coefficients (Lemma 1)
for $k = 0$ **to** $N-1$ **do**
  $\widehat{E}_k \leftarrow (Q_k + cI)^{-1}$; $\widehat{F}_k \leftarrow \widehat{E}_k A_k^\top$
  $\widehat{G}_k \leftarrow A_k\widehat{E}_k A_k^\top + B_k R_k^{-1} B_k^\top$
  $M_k \leftarrow I - c\widehat{G}_k$; $r_k^a \leftarrow \mathrm{rcond}(M_k)$ ▷ (a)
  $F'_k \leftarrow \widehat{F}_k M_k^{-1}$; $G'_k \leftarrow \widehat{G}_k M_k^{-1}$
  $E'_k \leftarrow \widehat{E}_k + cF'_k\widehat{F}_k^\top$
end for
Composition
$(\bar{E}_0, \bar{F}_0, \bar{G}_0) \leftarrow (E'_0, F'_0, G'_0)$; $s_1 \leftarrow r_0^a$
for $k = 1$ **to** $N-1$ **do**
  $r_k^b \leftarrow \mathrm{rcond}(E'_k + \bar{G}_{k-1})$ ▷ (b)
  update $(\bar{E}_k, \bar{F}_k, \bar{G}_k)$ by (23)–(26)
  $s_{k+1} \leftarrow \min(s_k, r_k^a, r_k^b)$
end for
Queries and certification
$\mathcal{C} \leftarrow \emptyset$
for $t = 1$ **to** N **do**
  $H_t \leftarrow (Q_T^{(t)} + cI)^{-1} + \bar{G}_{t-1}$; $r_t^c \leftarrow \mathrm{rcond}(H_t)$ ▷ (c)
  $\widehat{P}_0 \leftarrow \bar{E}_{t-1} - \bar{F}_{t-1}H_t^{-1}\bar{F}_{t-1}^\top$
  $J_t \leftarrow \frac{1}{2}x_0^\top(\widehat{P}_0^{-1} - cI)x_0 + wt$
  if $\min(s_t, r_t^c) \ge \tau$ **then** $\mathcal{C} \leftarrow \mathcal{C} \cup \{t\}$
end for

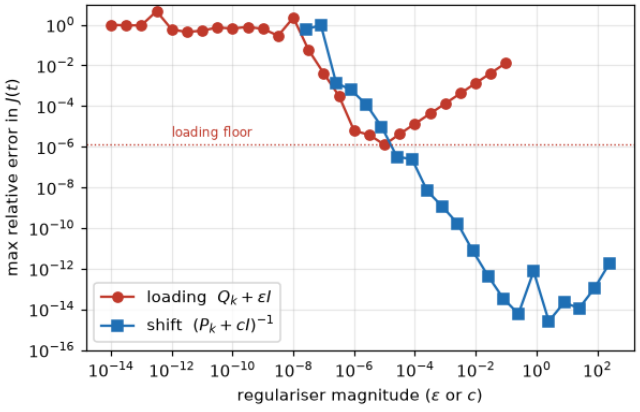


Fig. 1. Left: relative error against regularizer magnitude ($n = 6$, $N = 18$); loading is U-shaped with a floor near 10^{-6}, the shifted chart is flat over a wide range of c. Right: best-of-grid error against the coupling of an unpenalized direction ($n = 6$, $N = 30$, median of 4 seeds).

A. Perturbing Q_k

Diagonal loading restores invertibility by changing the problem. Fig. 1 (left) compares both methods at their best parameter: loading reaches 2.7×10^{-6} at $\varepsilon = 1.4 \times 10^{-6}$, while the shifted chart reaches 4.3×10^{-15} at its best c and 1.3×10^{-14} at c_0. Too small an ε amplifies $(Q_k + \varepsilon I)^{-1}$ through the composition; too large an ε solves a different problem. Across four configurations ($n \in \{4, 6, 8\}$, $N \in \{20, 60\}$, 8 seeds each) the shifted chart wins all 32 trials, at its best c and at c_0 alike, with median error ratios 2.7–5.1×10^{8}.

The floor depends on how the dynamics couple unpenalized and penalized states. Let $\mathcal{N}(Q_k) = \mathrm{span}\{v\}$, $\|v\| = 1$, and let $\kappa = \|A_k v - (v^\top A_k v)v\|$ be the part of $A_k v$ that leaves this direction. As κ grows from 0 to 2.17, loading's

TABLE I

HORIZON SELECTION AND COST ERROR WITHOUT TUNING. RANDOM: 200 INSTANCES × 5 DRAWS OF x_0 (WRONG OUT OF 1000, MAX. RELATIVE ERROR). FREE MODE: $\rho(A_k) = 1.05$ PLUS AN UNSTABLE, UNCONTROLLABLE, UNPENALIZED MODE, $n \in \{6, 12, 24\}$ (WRONG OUT OF 15).

	Random		Free mode	
Method	wrong	max err	N=120	N≥500
Loading, $\varepsilon = 10^{-4}$	0	3.4×10^{-2}	0	0
Loading, $\varepsilon = 10^{-6}$	0	3.5×10^{-4}	0	15
Loading, $\varepsilon = 10^{-8}$	46	3.9×10^{2}	2	2
Loading, $\varepsilon = 10^{-10}$	221	2.5×10^{3}	15	15
Loading, oracle ε	0	1.3×10^{-4}	0	11
Naive HOP	240	1.0×10^{2}	13	13
Shifted, $c_0 = 1$	0	1.0×10^{-11}	0	0†

†Horizons beyond $t \approx 215$–231 uncertified; $T^\star$ certified on all 15.

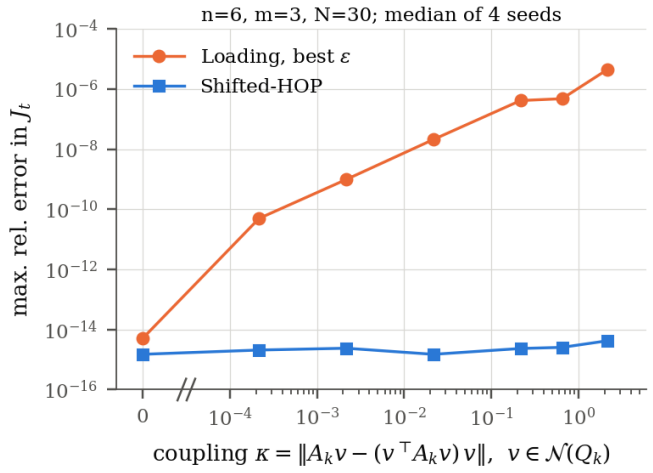


Fig. 2. Best-of-grid relative cost error against the coupling κ of an unpenalized direction ($n = 6$, $m = 3$, $N = 30$, median of 4 seeds).

best error rises from 5.0×10^{-15} to 4.6×10^{-6}, while Shifted-HOP stays at or below 4.2×10^{-15} (Fig. 2).

In practice ε must be fixed without a reference. Table I tests four fixed values on random instances and on *free-mode* instances, which add an unstable, uncontrollable, unpenalized mode. No fixed ε is both accurate and reliable: $\varepsilon \le 10^{-8}$ picks the wrong horizon on up to 22% of random trials, $\varepsilon = 10^{-6}$ on all 15 free-mode instances with $N \ge 500$, and only $\varepsilon = 10^{-4}$ is correct on both sets, at cost errors up to 1.4×10^{10}. Adaptive jitter is wrong on 24% of random trials, and on 24.7% when restricted to $T \ge 2$. Shifted-HOP with $c_0 = 1$ is correct on every instance of both sets.

B. Admissibility in Practice

The checks matter only when a pivot degrades, and they turn a silent failure into a reported one. On the free-mode instances of Table I, the reciprocal condition number of pivot (c) decays with the horizon to 10^{-18}. Without the checks, Algorithm 1 returns every cost anyway, with relative errors of 4×10^5 at $N = 500$ and 4×10^{13} at $N = 1000$, and selects the wrong horizon on 2 and 7 of 15 instances. With the checks, horizons beyond $t \approx 215$–231 are reported as uncertified, the true $T^\star$ is certified on every instance, and the error on certified horizons is at most 1.5×10^{-5}. On all other runs (346), every horizon was certified without a retry; the smallest reciprocal condition numbers were 2.3×10^{-7}, 3.0×10^{-8} and 1.9×10^{-7} for pivots (a), (b) and (c).

TABLE II

MEDIAN RUNTIME (MS), $n = 6$, $m = 2$, 15 REPEATS; SHIFTED-HOP INCLUDES c_0 AND THE ADMISSIBILITY CHECKS.

N	BF	Shifted	Naive	Loading	HOP (ref)
10	0.99	1.10	1.12	0.73	2.17
40	14.7	4.37	4.97	3.32	10.2
120	121	12.2	13.0	8.64	27.0

C. The Released HOP Implementation

The released HOP-DDP code [1] does not jitter Q_k; it builds an augmented $(n+1)$-dimensional system per stage and jitters that matrix. On a double integrator ($Q_k = \mathrm{diag}(1, 0)$), a segway and a cart with time-varying mass, all with singular Q_k, it selects the correct horizon on every trial when Q_T shares the running cost's null space. With a full-rank terminal cost, and the search over all $T \in \{1, \dots, N\}$ as in problem (5), it is wrong on 24.5–34.5% of trials; every wrong pick is $T^\star \le 2$, and restricting the search to $T \ge 2$ removes all failures. These counts are sensitive to the last bit of x_0: perturbing x_0 by one to four ulp changes HOP (REF)'s selection on 175 of 200 segway trials and moves its failure count between 48 and 69, while brute force changes on none. Shifted-HOP, Naive HOP and Loading are correct on every trial of this protocol. The authors' own examples search $T \ge 10$ and are unaffected.

D. Exactness and Scaling

On the random instances, Shifted-HOP and an independent condensed-QP formulation agree with brute force on every horizon choice; the median over x_0 draws of the worst cost error is 2.4×10^{-12} for Shifted-HOP and 1.8×10^{-14} for the QP. The true optimum is strictly interior in 38% of trials. The identities of Lemma 1 and Theorem 2 hold to within 1.5×10^{-11} on 200 singular instances. Error does not grow with the horizon: up to $N = 1000$ it stays below 10^{-12} on contracting random dynamics and below 1.7×10^{-10} on the open-loop unstable segway ($\rho(A) = 1.34$) and on random dynamics with $\rho(A_k) = 1.05$.

A fit $t = aN^p + b$ over $N \in \{10, \dots, 120\}$ gives $p = 1.98$ for brute force and $p \le 1.0$ for every composition method (Table II). Shifted-HOP runs at 0.88–0.98× Naive HOP's time and 1.3–1.5× Loading's; the admissibility checks account for 26–29% of its runtime at $N = 120$. HOP (REF) is 2.0–2.3× slower than Shifted-HOP. On the three physical systems at $N = 30$, Shifted-HOP is 2.6–2.8× faster than brute force.

V. CONCLUSION

We introduced a shifted inverse coordinate for horizon-optimal time-varying LQR that admits positive semidefinite

state costs, and showed that it can be absorbed into the inverse-coordinate LFT composition while retaining $O(Nn^3)$ horizon search and, in exact arithmetic, exact recovery of the Riccati solution. Admissibility of the shifted stage maps holds for all small c when the dynamics contract the unpenalized directions; otherwise it fails for at most finitely many c outside a degenerate case. A run-time check certifies each horizon from the existing factorizations and reports uncertified horizons instead of returning their costs. In our experiments, every horizon was certified on the random and physical test problems. On instances with an unstable, uncontrollable, unpenalized mode, long horizons were uncertified while the optimal horizon remained certified. With the checks enabled, the method ran at the speed of an adaptive-jitter inverse coordinate and slower than diagonal loading.

The analysis is limited to linear time-varying LQR with positive semidefinite costs. Certification bounds the conditioning of each pivot, not the final error. Indefinite costs, as arise from nonconvex running costs in DDP, additionally require $P_k \succ -cI$, which a fixed c does not guarantee. Embedding the construction in DDP for nonlinear systems, and extending it to constrained problems [16], are left to future work.

ACKNOWLEDGEMENT

The authors acknowledge the use of LLMs for improving the grammatical structure and clarity of this manuscript.

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